

Hadisa Abroo

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RESEARCH PROFILE

Researcher in applied and theoretical quantum physics working on non-Hermitian quantum mechanics and exceptional-point-enhanced sensing, coherent-state quantum metrology under decoherence, and coherence-assisted entanglement dynamics in semiconductor quantum dot and photonic platforms, with quantum Fisher information, Schmidt decomposition, and von Neumann entropy as the principal analytical tools. First author on five quantum-theory manuscripts, one published in *Physica B: Condensed Matter* (Elsevier) and one sole-authored. Completed quantum technologies training at the School of Natural Sciences, NUST. Experienced in analytical quantum theory, Lindblad master equation modelling, and quantum circuit simulation via IBM Qiskit.

Keywords Quantum sensing · Quantum metrology · Quantum entanglement · Non-Hermitian physics

EDUCATION

MPhil in Physics · Riphah International University, Pakistan CGPA 3.38 / 4.0 Specialisation: Quantum Coherence, Entanglement & Quantum Entropy	2022 – 2025
MSc in Applied Physics · Riphah International University, Pakistan CGPA 3.53 / 4.0 Focus: Quantum Mechanics, Electromagnetic Theory, Mathematical Physics	2020 – 2022
BSc in Mathematics & Physics (Double Major) · University of Sargodha, Pakistan First Division Core: Classical Mechanics, Optics, Linear Algebra, Calculus	2017 – 2019

PUBLICATIONS AND PREPRINTS

*Corresponding author. † Manuscript available upon request.

PUBLISHED

- [1] Abroo, H., & Iqbal Bashir, A.* “Quantum Coherence-Assisted Quantum Dot–Photon Entanglement via Schmidt Decomposition.” *Physica B: Condensed Matter* (Elsevier), 2026. Available online 11 September 2026. [ScienceDirect](#)

UNDER REVIEW

- [2] Abroo, H., Bashir, S., & Iqbal, A.* “Tunable Photon–Graphene Quantum Dots Entanglement by von Neumann Quantum Entropy.” Submitted to a peer-reviewed journal, 2026.†
- [3] Abroo, H., & Iqbal, A.* “Field-Tunable Photon–Quantum-Dot Entanglement in a Λ -Type GaAs/AlGaAs Quantum Dot Molecule via Coherent Population Trapping and Quantum Entropy.” Submitted to a peer-reviewed journal, 2026. Companion to the cascade (Ξ) study [1].†

PREPRINTS

- [4] Abroo, H. “Exceptional Points in Non-Hermitian Quantum Systems and Their Role in Enhanced Sensing: A Unified Analytical Framework for Signal, Open-System Noise, and Detectivity.” Sole-authored preprint, 2026 (arXiv:quant-ph, submission in progress).†

IN PREPARATION

- [5] Abroo, H., Bibi, T., & Iqbal, S.* “Quantum Fisher Information of Gazeau–Klauder Coherent States under Markovian Decoherence.” Manuscript in preparation, 2026.

RESEARCH EXPERIENCE AND TRAINING

Quantum Internship 2026 · School of Natural Sciences, NUST, Islamabad Jun – Aug 2026
Supervisor: Prof. Dr. Shahid Iqbal, Department of Physics & Astronomy

- Completed an eight-week in-person programme in quantum technologies, combining lectures with laboratory sessions.
- Quantum circuit design and multi-qubit entanglement generation; quantum communication, cryptography, and teleportation protocols; quantum algorithms and complexity advantage.
- Hands-on implementation in Qiskit and QuTiP; capstone quantum programming project.

Exceptional-Point-Enhanced Quantum Sensing 2025 – 2026

- Analytical treatment of non-Hermitian Hamiltonians; eigenvalue and exceptional-point analysis.
- Sensitivity enhancement characterised via quantum Fisher information; unified analytical framework deriving signal and noise from a single parametrised Hamiltonian.
- PT-symmetry breaking, spectral topology, and connection to quantum metrology protocols.
- Mathematica symbolic simulations; publication-quality figure preparation.

Quantum Coherence-Assisted Entanglement via Quantum Entropy (MPhil Thesis) 2022 – 2025

- Theoretical framework for coherence as a resource for entanglement generation in quantum materials.
- Entanglement quantified by Schmidt decomposition in GaAs/AlGaAs quantum dot systems; entanglement reaches one full ebit at control-laser resonance and is stabilised by increasing control power.

- Density matrix evolution modelled via Lindblad master equation; concurrence, von Neumann entropy, and quantum discord as diagnostics, with IBM Qiskit simulations under decoherence.
- Systematic review of 80+ papers across quantum thermodynamics, communication, and open systems.

Photon–Quantum-Dot Entanglement in Coupled and Hybrid Platforms

2025 – 2026

- Entanglement characterisation in Λ -type GaAs/AlGaAs quantum dot molecules and photon–graphene quantum dot hybrid systems.
- Coherent population trapping, dark states, and electromagnetically induced transparency as control mechanisms; von Neumann entropy as the primary diagnostic.
- Collaborative theoretical work; two manuscripts under journal review.

OPEN-SOURCE RESEARCH CODE

Repository: github.com/hadisaabroo99-pixel

Project	Content	Status
Quantum-Enhanced Metrology: Ramsey Interferometry, Squeezed States & Standard Quantum Limit	Heisenberg-limit estimation, quantum Fisher information, Ramsey schemes, squeezed-state metrology, EP-enhanced sensing. Qiskit + Mathematica. Supports [4], [5].	Active
Quantum Entanglement & Non-local Correlations: Bell States, CHSH Violation & Decoherence Dynamics	Bell-state preparation and measurement, CHSH violation, teleportation, entanglement monotones, Bloch sphere visualisation, decoherence simulations. Qiskit + Python. Supports [1]–[3].	Active
Quantum Computing Algorithms: Grover Search, Quantum Error Correction & Entanglement Scaling	Grover's algorithm, surface and repetition code QEC (stabilizer formalism), entanglement entropy scaling. Qiskit.	In progress

TECHNICAL SKILLS

Quantum Software	IBM Qiskit, QuTiP, Python (quantum simulation)
Scientific Computing	NumPy, SciPy, Matplotlib, Pandas
Symbolic Computation	Mathematica / Wolfram Language
Scientific Writing	$\LaTeX$ (Overleaf), Zotero
Design & Visualisation	Adobe Creative Suite (research figures, posters)

RESEARCH INTERESTS

Quantum information theory · Quantum sensing and metrology · Quantum entanglement and coherence · Non-Hermitian quantum systems · Open quantum systems and thermodynamics · Quantum communication and QKD

SELECTED ACHIEVEMENTS

- Five quantum-theory manuscripts as first author: one published in *Physica B: Condensed Matter* (Elsevier), two under journal review, one sole-authored preprint, and one in preparation.
- Two public open-source quantum simulation repositories on GitHub, with a third in progress.
- MSc CGPA 3.53 / 4.0 (top cohort); BSc First Division.
- IELTS Academic Band 7.0 (C1 — English proficiency).

CONFERENCES AND WORKSHOPS

International Conference on Emerging Quantum Sciences & Technologies (ICEQST 2026)

27 Aug 2026

Attendee · School of Natural Sciences, NUST, Islamabad.

CERTIFICATIONS AND LANGUAGES

Practical Quantum Computing with IBM Qiskit, Packt (UK)			2023
Quantum Mechanics, University of Colorado Boulder (Coursera)			2023
How to Write and Publish a Scientific Paper, École Polytechnique (Coursera)			2022
Scientific Writing with $\text{\LaTeX}$ & Overleaf, Fred Hutchinson Cancer Center (Coursera)			2022
English	C1 — Advanced	IELTS Academic 7.0	
German	A1 — Elementary	Actively developing	
Urdu	C2 — Native		

ADDITIONAL EXPERIENCE

Research Visualisation Designer · Sixdeees

2021 – present

Scientific poster design, data visualisation, and research communication graphics.

Physics & Mathematics Instructor · ZA Public Schools / Allied School

2018 – 2020

Secondary-level instruction; curriculum development; science communication.

References Available on request, from Riphah International University and the School of Natural Sciences, NUST.